

A Hybrid AHP–TOPSIS Approach to Laptop Selection for Office Workers in Vietnam

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Abstract

[Abstract to be written in Phase 6 — gap → method (AHP–TOPSIS, three stages) → results (weights, ranking) → implication.]

Keywords: Multi-Criteria Decision-Making; AHP; TOPSIS; laptop selection; office workers; Vietnam.

1 Introduction

The laptop has become a basic working tool across most occupations, and its consumption has expanded steadily. Sönmez Çakır and Pekkaya (2020) report that 161.6 million laptops were sold in 2017, and Gaur (2023) record a rise to 218 million units in 2020 and 225 million in 2021. As the catalogue fragments into many brands, models, and configurations, buyers must compare technical attributes expressed on different scales that frequently conflict, which places the choice within the scope of Multi-Criteria Decision-Making (MCDM). Among MCDM techniques, the Analytic Hierarchy Process (AHP) and the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) are among the most widely used (Basílio et al., 2022; Zyoud & Fuchs-Hanusch, 2017), and their combination recurs in applied research (Tighnavard Balasbaneh et al., 2025).

[Phase 6 — condense problem statement, gap, contributions, and research questions; see SUON-NOI-DUNG.md §P1.]

2 Literature Review

[Phase 6 — see SUON-NOI-DUNG.md §P2.]

3 Methodology

3.1 Research Framework and Criteria

[Phase 6 — three-stage framework and the fourteen-criteria table; see SUON-NOI-DUNG.md §P3.]

3.2 AHP for Criteria Weighting

Criteria weights are derived through AHP. Expert pairwise judgements on the Saaty 1–9 scale form a reciprocal matrix $A = [a_{ij}]$; the weight of each criterion is the normalised row average

$$w_i = \frac{1}{n} \sum_{j=1}^n \frac{a_{ij}}{\sum_{k=1}^n a_{kj}}, \quad (1)$$

and consistency is verified through $CR = CI/RI$ with

$$CI = \frac{\lambda_{\max} - n}{n - 1}, \quad CR \leq 0.10. \quad (2)$$

3.3 TOPSIS for Ranking

Given the weights, the candidate laptops are ranked with TOPSIS. The decision matrix is vector-normalised, $r_{ij} = x_{ij} / \sqrt{\sum_i x_{ij}^2}$, and weighted, $v_{ij} = w_j r_{ij}$. After forming the positive and negative ideal solutions A^+ and A^- , the separation measures S_i^+ and S_i^- yield the closeness coefficient

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}, \quad (3)$$

and alternatives are ranked in descending order of C_i .

4 Results

[Phase 4-5 output — see *SUON-NOI-DUNG.md* §P4.]

5 Discussion

[Phase 6 — see *SUON-NOI-DUNG.md* §P5.]

6 Conclusion

[Phase 6 — see *SUON-NOI-DUNG.md* §P6.]

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